Venkata Lakshmi Parimala Pasupuleti

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AgentProbe: github.com/venkatalakshmiparimala/AgentProbe | Virginia, USA

PROFESSIONAL SUMMARY

Staff AI engineer with **production healthcare NLP experience** and end-to-end LLM/agentic systems ownership. Built **AI on FHIR** - HIPAA-compliant clinical NLP pipeline (BERT, spaCy, embeddings) mapping natural language queries to FHIR APIs, cutting retrieval time by ~60%. Built **AgentProbe** - production LangGraph agentic platform with multi-provider LLM routing (Anthropic, OpenAI, Groq), ChromaDB vector search, and CI/CD-gated evaluation engine. 8+ years Python production engineering at Entrata (Docker, AWS S3/EC2/RDS, 1.6M records, 100K+ users). M.S. Data Analytics Engineering, GMU (GPA 3.87, May 2026). *github.com/venkatalakshmiparimala/AgentProbe*

CORE COMPETENCIES

Clinical NLP & Healthcare AI (FHIR) | LLM Integration & Agentic Systems | Production ML Lifecycle (Deploy → Monitor → Eval)

Python (8+ years production) | NLP: BERT, spaCy, Embeddings, Hugging Face | Vector Search (ChromaDB)

CI/CD for ML (GitHub Actions) | Docker + AWS (S3, EC2, RDS, Glue) | FastAPI / REST APIs | LLM Evaluation & Observability

HIPAA Compliance & PHI Security | PyTorch / scikit-learn / TensorFlow

EXPERIENCE

Entrata India (Db Xento Systems)

Jan 2017 - Aug 2024

Software Engineer → Associate Software Engineer → Trainee

Pune, India

- Applied **Python, PyTorch, and OpenAI API** to automate domain-specific data transformation across **1.6M financial records** - production AI on regulated data requiring careful validation and output monitoring.
- Led **Docker-based containerization** for 30%+ of microservices; cut execution time by **50%**; maintained **99.5% uptime** via TDD/BDD - the containerized, reproducible environments ML production systems require.
- Achieved **100% PHI compliance** migrating client data to AWS S3 with Libsodium-encrypted pipelines - zero security incidents across 100K+ users; directly applicable to HIPAA-sensitive clinical AI systems.
- Improved production query performance by **30%** across 200+ SQL queries; built **5+ internal debugging tools** reducing resolution time by **40%**; mentored engineers through code reviews and documentation.
- Built and delivered **20+ full-stack modules** (Python, JavaScript, PostgreSQL) across **120+ Agile sprints** for **100K+ monthly active users**.

George Mason University

Aug 2025 - May 2026

Graduate Teaching Assistant - Python & Databases

Fairfax, Virginia

- Mentored 30+ students in Python and databases; enforced quality standards and drove iterative feedback loops - same discipline as ML system quality ownership.

PROJECTS

AI on FHIR: Clinical NLP Pipeline for Medical Records

Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings · NumPy · Pandas

NLP system mapping natural language queries to FHIR APIs for clinical record retrieval - directly in IMO Health's domain of unstructured clinical data and healthcare terminology. **Clinical NLP stack**: BERT + spaCy + embeddings for entity extraction, classification, and semantic search across medical record fields. **HIPAA-compliant RBAC**: role-based access controls for PHI - same compliance model IMO operates under. **Production deployment**: Flask + React, containerized with Docker, EN/ES multilingual support. **~60% retrieval time reduction.** *github.com/venkatalakshmiparimala/AI_on_FHIR*

AgentProbe: Production LLM + Agentic AI Evaluation Platform OPEN SOURCE

Python · LangChain · LangGraph · FastAPI · ChromaDB · React · Anthropic · OpenAI · Groq · Ollama · Docker · GitHub Actions CI/CD

End-to-end ownership of a production LLM/agentic AI platform - the full ML lifecycle from design to CI/CD-gated deployment. **LLM integration**: multi-provider routing across Anthropic, OpenAI, Groq, Ollama - dynamic selection by task, latency, and capability. **Agentic systems**: 3-agent LangGraph runtime (Orchestrator, RAG Tracer, Auto Fix) with multi-step reasoning and tool calls. **Vector search**: ChromaDB + embeddings + retrieval quality scoring. **CI/CD eval pipeline**: 5-metric LLM-as-judge engine (factual accuracy, hallucination, retrieval, completeness, compliance) with regression tracking and build gates - automated monitoring that catches quality drift before it reaches users. **Observability**: 15-endpoint FastAPI backend + React dashboard. Open-source. *github.com/venkatalakshmiparimala/AgentProbe*

CrashClarity: Predictive ML Pipeline - 85%+ Accuracy on 100K+ Records

Python · R · scikit-learn · Random Forest · Logistic Regression · Feature Engineering · GitHub Pages

Full ML lifecycle: feature engineering → model selection → evaluation (MDI + Permutation Importance) → stakeholder dashboard. **85%+ accuracy.**

EDUCATION

M.S. Data Analytics Engineering - George Mason University 2024-2026

GPA: 3.87 · Virginia, USA · Statistical modeling, ML systems, experimental design, production data pipelines

B.Tech Computer Science & Engineering - Sree Venkateshwara College of Engineering 2013-2017

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, R, Bash, HTML/CSS

AI & LLM: LangChain, LangGraph, LLM APIs (Anthropic, OpenAI, Groq, Ollama), Agentic Systems, Prompt Engineering, RAG Pipelines, ChromaDB, LLM-as-Judge Evaluation, LLM Monitoring

NLP & ML: BERT, spaCy, Hugging Face, PyTorch, TensorFlow, scikit-learn, Embeddings, Feature Engineering, Random Forest, Statistical Modeling, NumPy, Pandas

Healthcare & Compliance: FHIR APIs, HIPAA-compliant RBAC, PHI Encryption (Libsodium), Secure Coding Standards, Clinical NLP

Cloud & DevOps: AWS (S3, EC2, RDS, Glue), Azure, Docker, CI/CD (GitHub Actions), FastAPI, Flask, React, PostgreSQL, MySQL

Process: TDD/BDD, Integration Testing, Open-Source Development, Git, JIRA, Agile/Scrum